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Baseline (SE(3) trajectory)

# Data Format

- The data collection path can be found in the Download pages of our website. All data were recorded using the ROS timestamp. All data are provided in a compressed 'tar' format and stored in a folder with the date and time the data was collected.

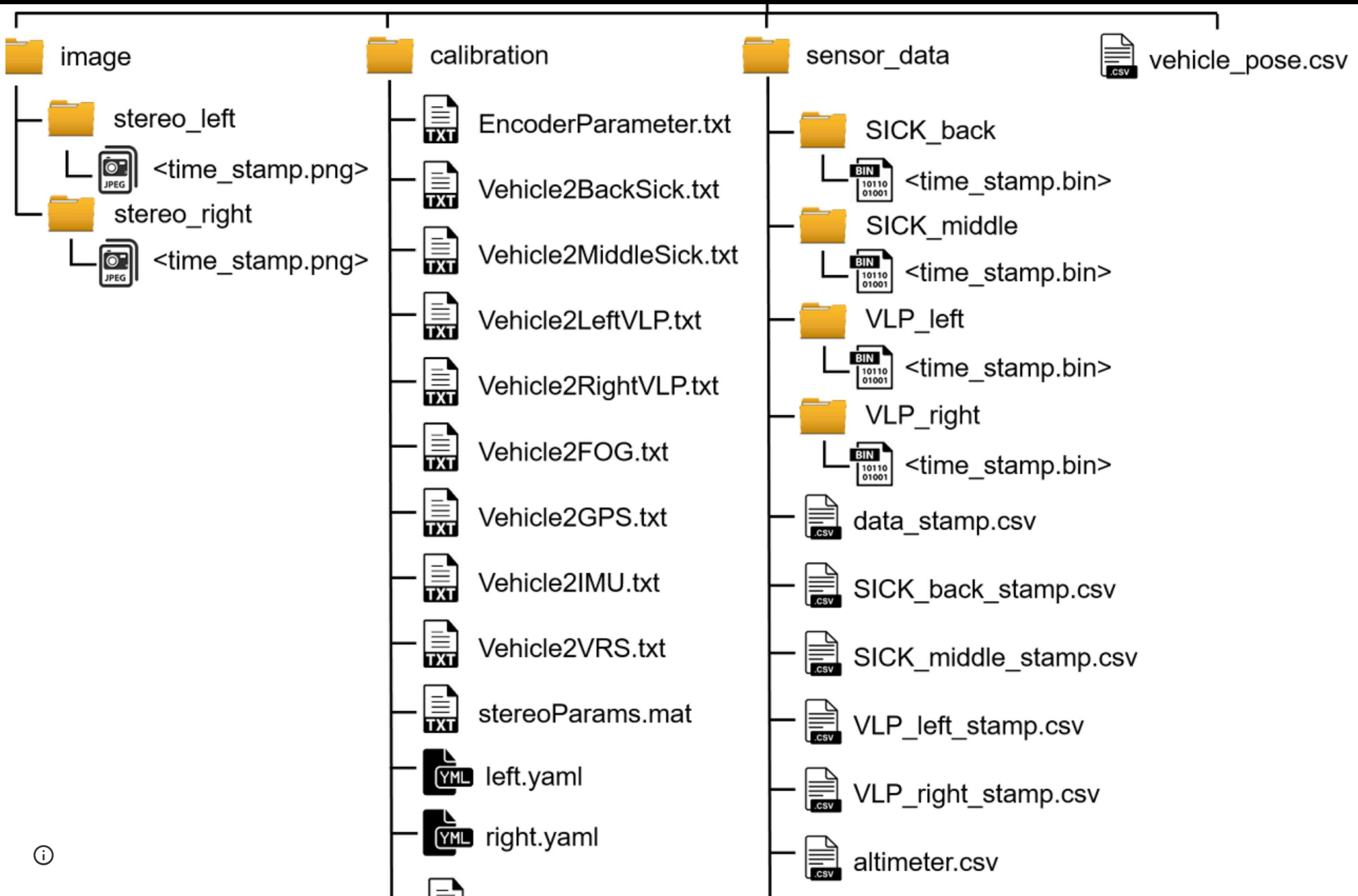
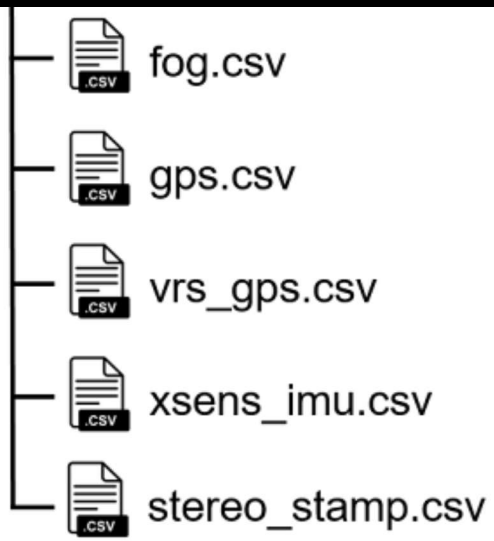


image
stereo_left
<time_stamp.png>
stereo_right
<time_stamp.png>
calibration
EncoderParameter.txt
Vehicle2BackSick.txt
Vehicle2MiddleSick.txt
Vehicle2LeftVLP.txt
Vehicle2RightVLP.txt
Vehicle2FOG.txt
Vehicle2GPS.txt
Vehicle2IMU.txt
Vehicle2VRS.txt
stereoParams.mat
left.yaml
right.yaml
sensor_data
SICK_back
<time_stamp.bin>
SICK_middle
<time_stamp.bin>
VLP_left
<time_stamp.bin>
VLP_right
<time_stamp.bin>
data_stamp.csv
SICK_back_stamp.csv
SICK_middle_stamp.csv
VLP_left_stamp.csv
VLP_right_stamp.csv
altimeter.csv
vehicle_pose.csv


fog.csv
gps.csv
vrs_gps.csv
xsens_imu.csv
stereo_stamp.csv

### 1. 3D LiDAR Data

- The timestamp of the last packet is used as the timestamp of the data at the end of one rotation. 3D LiDAR data is stored in the 'VLP_left' and 'VLP_right' folders in the sensor_data folder as floating point binary format, and the timestamp of each rotation data is the name of the file ('time_stamp.bin'). The timestamps of all 3D LiDAR data are stored sequentially in 'VLP_left_stamp.csv' and 'VLP_right_stamp.csv'.
  - **[x, y, z, R] : x, y and z denote the local 3D Cartesian coordinate of each LiDAR sensor. R is the reflectance value.**

### 2. 2D LiDAR Data

- The 2D LiDAR data are stored in the 'SICK_back' and 'SICK_middle' folder in the sensor_data folder as floating point binary format. Similar to 3D LiDAR data, the timestamp of each scan data is the name of the file. The timestamps of all 2D LiDAR data are stored sequentially in 'SICK_back_stamp.csv' and 'SICK_middle_stamp.csv'
  - **[r, R] : To reduce the file size, the data of 2D LiDAR consists of two items. r is the range value of each point, and R is the reflectance value. The sensor's Field of View (FOV) is 190°. The start angle of the first data is −5°, and the end angle is 185°. The angle difference between each sequential data is 0.666°. Each point can be converted to a Cartesian coordinate using this information. Please refer to the paper for more details.**

## Stereo Camera Data

- The stereo images were acquired at 10 Hz and stored in the lossless PNG format in unrectified 8-bit Bayer pattern images. The Bayer pattern of the images is RGGB. An external trigger was used for accurate synchronization of the stereo camera. Images are stored in the 'stereo_left' and 'stereo_right' folders in the 'image' folder and each image is named using timestamps. The timestamps of all stereo images are stored in order in 'stereo_stamp.csv' located in the 'sensor_data' folder.

- Note: Ver 1 means the Download (LiDAR) tab, and Ver 2 means the Download (LiDAR + Stereo) tab.

## 1. GPS (Global Positioning System)

- The 'sensor_data/gps.csv' file stores the global position measured by commercial level GPS sensor.
  - **[timestamp, latitude, longitude, altitude, 9-tuple vector (position covariance)]**

## 2. VRS (Virtual Reference Station) GPS

- The 'sensor_data/vrs_gps.csv' file stores the accurate global position measured by VRS GPS sensor.
  - **Ver 1: [timestamp, latitude, longitude, x coordinate, y coordinate, altitude, fix state, number of satellite, horizontal precision, latitude std, longitude std, altitude std, heading validate flag , magnetic global heading, speed in knot, speed in km, GNVTG mode]**
  - **Ver 2: [timestamp, latitude, longitude, x coordinate, y coordinate, altitude, fix state, number of satellite, horizontal precision, latitude std, longitude std, altitude std, heading validate flag , magnetic global heading, speed in knot, speed in km, GNVTG mode, ortometric altitude]**
    - where the x and y coordinates are in the UTM coordinate system of the meter unit. The fix state is a number indicating the state of the VRS GPS. For example, 4, 5, and 1 indicates the fix, float, and normal states, respectively. The accuracy of VRS GPS in the sensor specification is the value at the fix state.

## 3. IMU (Inertial Measurement Unit)

- The 'sensor_data/imu.csv ' file stores the incremental rotational pose, gyro, acceleration, magnet filed data measured by AHRS IMU sensor. Urban00-05 and campus00 have only quaternion and eular data. Data set after urban05 and campus00 contain gyro, acceleration, and magetfield data.
  - **Ver1: [timestamp, quaternion x, quaternion y, quaternion z, quaternion w, Euler x, Euler y, Euler z]**
  - **Ver2: [timestamp, quaternion x, quaternion y, quaternion z, quaternion w, Euler x, Euler y, Euler z, Gyro x, Gyro y, Gyro z, Acceleration x, Acceleration y, Acceleration z, MagnetField x, MagnetField y, MagnetField z]**

## 4. FOG (Fiber Optic Gyro)

- The 'sensor_data/fog.csv ' file stores the relative rotational motion between consecutive sensor data.

- The 'sensor_data/encoder.csv ' file stores the incremental pulse count values of wheel encoder.
  - **[timestamp, left count, right count]**

### 6. Altimeter

- The 'sensor_data/altitude.csv ' file stores the altitude values measured by the altimeter sensor.
  - **[timestamp, altitude]**

## Calibration Data

- Calibration is performed whenever data is acquired. Therefore, the calibration results of each sensor are provided in the 'calibration' folder along with each data set. The calibration data is provided in both the Euler format and SE(3) format.

## Sequence Data

- The 'sensor_data/data_stamp.csv' file stores the names and timestamps of all sensor data in order.
  - **[timestamp, sensor name]**

- A baseline is the trajectory of the vehicle generated by various algorithms. The generated baseline trajectory is stored in 'vehicle_pose.csv' at a rate of 100 Hz. However, it is not desirable to use baseline trajectory as the ground truth for mapping or localization benchmarking as the SLAM results depend on the complexity of the urban environment.
  - **[timestamp, P(0,0), P(0,1), P(0,2), P(0,3), P(1,0), P(1,1), P(1,2), P(1,3), P(2,0), P(2,1), P(2,2), P(2,3)]**
    - **where P is 4X4 transformation matrix at each time. Rotation matrix and translation matrix is stored in vector form.**

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